FloatSat Presentation

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The Floating Satellite System as an Educational Platform for Space Applications

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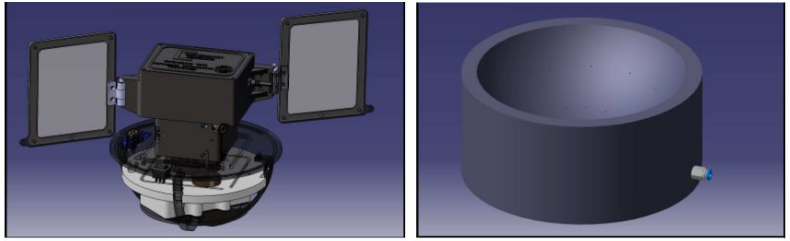
***Abstract*—In this paper, an approach is presented for building a small laboratory based Floating–Satellite(Float–Sat). This Float–Sat will serve as a test–bench for the students to learn basic satellite subsystems. This can also be used for the development and verification of different control algorithms and strategies for various kinds of space missions. This Float–Sat also provides a near–to a frictionless environment similar to space.**

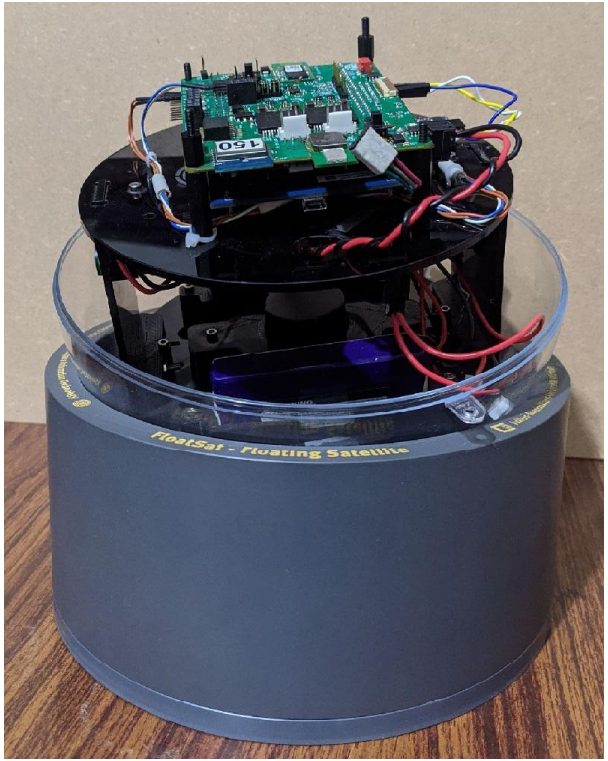
***Index Terms*—Aerospace, Real–time–operating–systems, Operational Modes of Satellites.**



a: FloatSat b: Floating Platform (SABU)

Fig. 1: The Floating Satellite (FloatSat) system

HARDWARE



Reaction wheel

Motor

1. Operating voltage: 1-12V
2. Speed: 11000 RPM at 12V
3. Encoder counts: 16 per revolutions

Reaction wheel

4. Mass: 298 grams
5. 1.175×10^{-4} Kgm^2

Fig. Motor and reaction wheel

OBC: STM32F4 Discovery Board

STM32F4

Arm® 32-bit Cortex®-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait state execution from Flash memory, memory protection unit, 210 DMIPS/ 1.25 DMIPS/MHz (Dhrystone 2.1), and DSP instructions

1. Frequency: Upto 168 MHz
2. 92+4 Kbytes of SRAM
3. 1 Mbyte of Flash memory
4. Mass: 90 gram

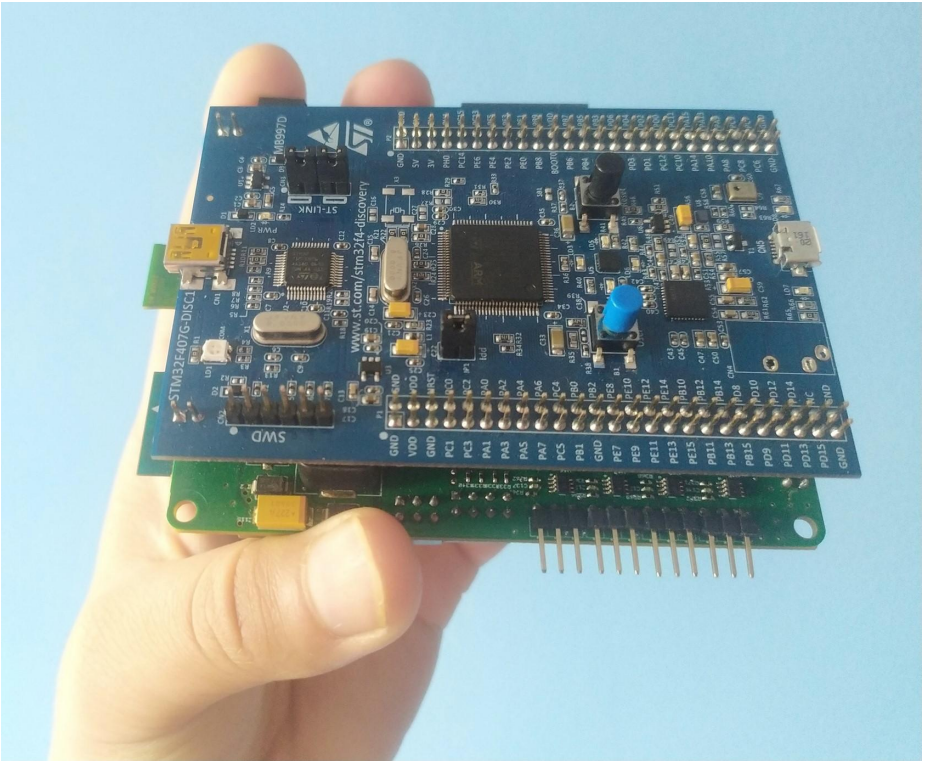


Fig. Discovery Board with addon shield

Add-on Board

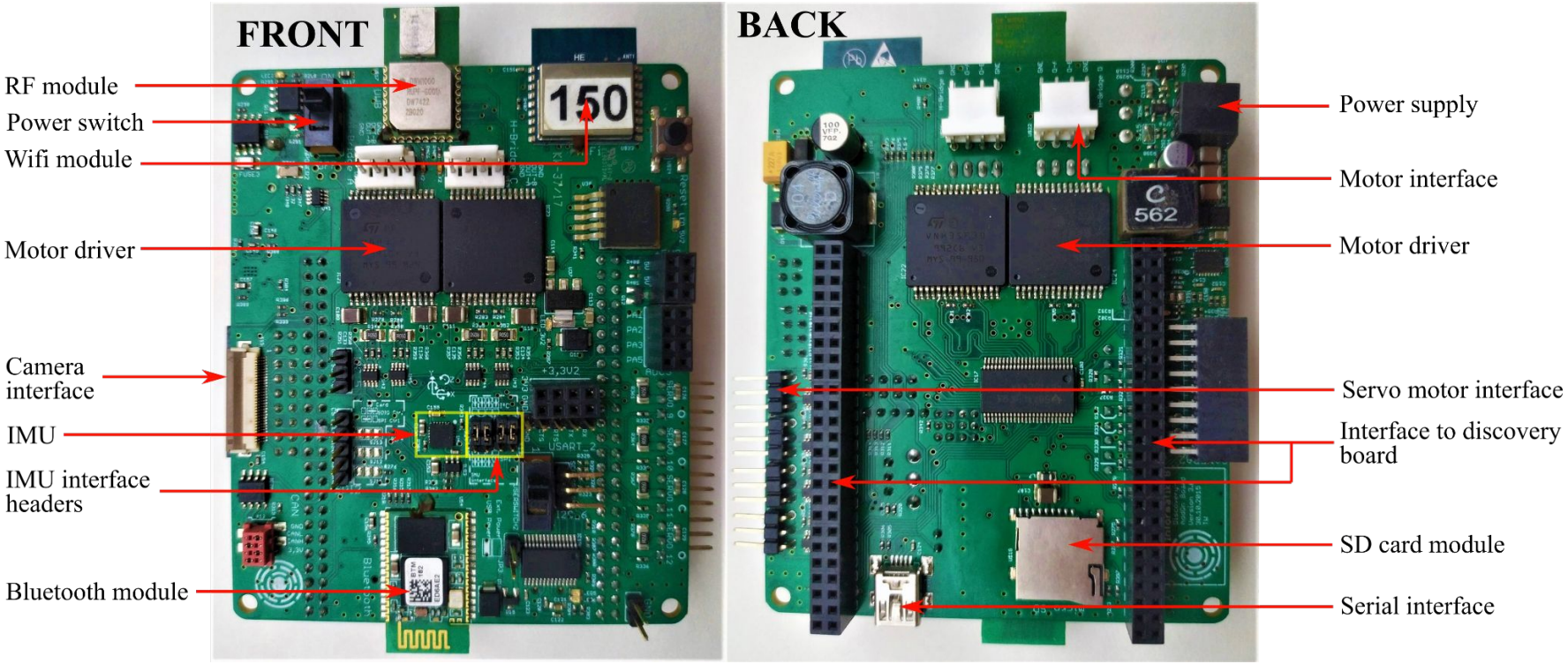


Fig. Add-on board where discovery board fits.

SOFTWARE

Realtime Onboard Dependable Operating System

RODOS

Real Time Kernel Design for Dependability

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Introduction

The NetworkCentric core avionics machine consists of several harmonised components which work together to implement dependable computing in a simple way.

***Repository*:** https://gitlab.com/rodos/rodos

RODOS

Features

1. Hardware abstraction
2. Timing abstraction
3. Preemptive dynamic schedule

Philosophy

1. Dependability*
2. Simplicity

* *Dependability* is the ability to provide services that can defensibly be trusted within a time-period.

System Diagram

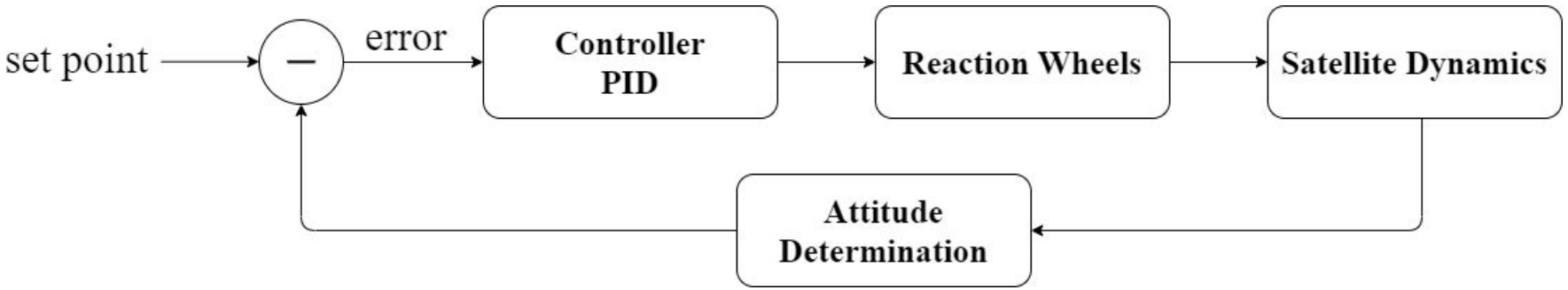

Fig. The high level diagram of attitude control system for FloatSat

Attitude Determination

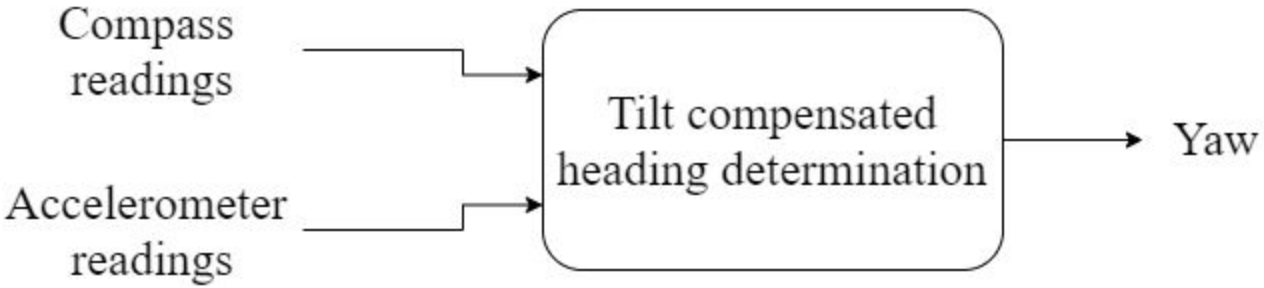

Fig. Tilt compensated yaw determination

Control System

$$u(t) = K_p * e(t) + K_i \int e(t)\,dt + K_d \frac{d}{dt} e(t)$$

where,

$u(t) = \text{output}$

$e(t) = \text{error} = \text{set point - feedback}$

$K_p,\ K_i,\ K_d = \text{PID gains}$

PID Laplace Transform

$$u_p(s) = K_p\, e(s)$$

$$u_i(s) = \frac{1}{s} K_i\, e(s)$$

$$u_d(s) = sK_d\, e(s)$$

$$u(s) = u_p(s) + u_i(s) + u_d(s)$$

Tustin's Approximation

$$s = \frac{2}{T}\frac{z-1}{z+1}$$

$$\begin{aligned} u[n] &= u_p[n] + u_i[n] + u_d[n] \\ &= K_p * e[n] + \\ &\quad \frac{1}{2}K_i T\left(e[n] + e[n-1]\right) + u_i[n-1] + \\ &\quad \frac{2}{T}K_d\left(e[n] - e[n-1]\right) - u_i[n-1] \end{aligned}$$

Fig. Discretized Equations of PID

DEMONSTRATION

Thank You !

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Team Members